

CREDIT CONSTRAINTS AND THE CYCLICALITY OF R&D INVESTMENT: EVIDENCE FROM FRANCE

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Abstract

We use a French firm-level data set containing 13,000 firms over the period 1994–2004 to analyze the relationship between credit constraints and firms' R&D behavior over the business cycle. Our main results can be summarized as follows: (i) R&D investment is countercyclical without credit constraints, but it becomes procyclical as firms face tighter credit constraints; (ii) this result is only observed for firms in sectors that depend more heavily upon external finance, or that are characterized by a low degree of asset tangibility; (iii) in more credit-constrained firms, R&D investment plummets during recessions but does not increase proportionally during upturns. (JEL: E22, E32, O16, O30, O32)

1. Introduction

A Schumpeterian view of business cycles and growth is that recessions provide a cleansing mechanism for correcting organizational inefficiencies and for encouraging firms to reorganize, innovate or reallocate to new markets. This cleansing effect of recessions is also to eliminate those firms that are unable to reorganize or innovate. Schumpeter (1942) himself would summarize that view as follows; “[Recessions] are but temporary. They are means to reconstruct each time the economic system on a more efficient plan.” This of course assumes that firms can always borrow enough

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funds to either reorganize their activities or move to new activities and markets. Investment choices are indeed dictated by an opportunity cost effect: namely, the opportunity cost of long-term innovative investments instead of short-term capital investments is lower in recessions than in booms.¹ Hence, long-term investments should be countercyclical, whereas short-term investments should be procyclical (see Hall 1993; Gali and Hammour 1992; Aghion and Saint-Paul 1998; Bean 1990; Bloom 2007).

However, this is only true to the extent that firms are not credit constrained. As emphasized by Aghion et al. (2010), henceforth AABM, things become quite different when credit market imperfections prevent firms from innovating and reorganizing in recessions. In particular, suppose that firms can choose between short-run capital investment and long-term R&D investment, that innovating requires that firms survive short-run liquidity shocks, and that to cover liquidity costs firms can rely only on their short-run earnings plus borrowing. Whenever the firm is hit by a bad (idiosyncratic or aggregate) shock, its current earnings are reduced, and therefore so is the firm's ability to borrow in order to innovate. This in turn implies that a negative shock should hit R&D investments and innovation more in firms that are more credit constrained. In other words, R&D investments should be expected to be less countercyclical in firms facing tighter credit constraints.

In this paper, we first present a simple model emphasizing those mechanisms and test its predictions using a French firm-level panel data set that contains information both, on the extent of credit constraints at the firm level each year, and on R&D investments by the firms. The firm-level database we use has been collected by the Banque de France. The sample includes about 13,000 firms (all of them having at least one year of positive R&D investment) and covers the period 1994–2004. It contains an important number of small and medium firms that are particularly prone to be hit by credit constraints, and are thus especially relevant for the study of the above-mentioned mechanisms.

We regress R&D investment on firm sales growth and its interaction with credit constraints.² Our main results can be summarized as follows: First, R&D investment is countercyclical without credit constraints, but becomes procyclical as firms face as firms face sufficiently tight credit constraints. Figure 1 provides a first illustration to this effect. It depicts the relationship between R&D and sales growth.³ In particular we see that credit constrained firms exhibit a clearly procyclical R&D investment pattern. Our second finding is that this effect of credit constraints is only observed

1. The underlying idea is that firms can use their human and physical assets either for current production or for reorganizing their activities or for innovating. If they experience a boom (with high current demand for their product or high current productivity), firms will rather choose to allocate their assets to current production at the expense of reorganization or innovation activities. And the converse is true if firms experience a downturn.

2. In a previous version of the paper (see Aghion et al. 2008), we used the ratio of R&D over total investment as a dependent variable; the results were qualitatively similar.

3. This figure shows the Lowess smooth curve of the relationship between R&D growth and sales growth for two groups of firms defined according to the degree of credit constraints they face (we detail and discuss our measure of credit constraints in Section 3).

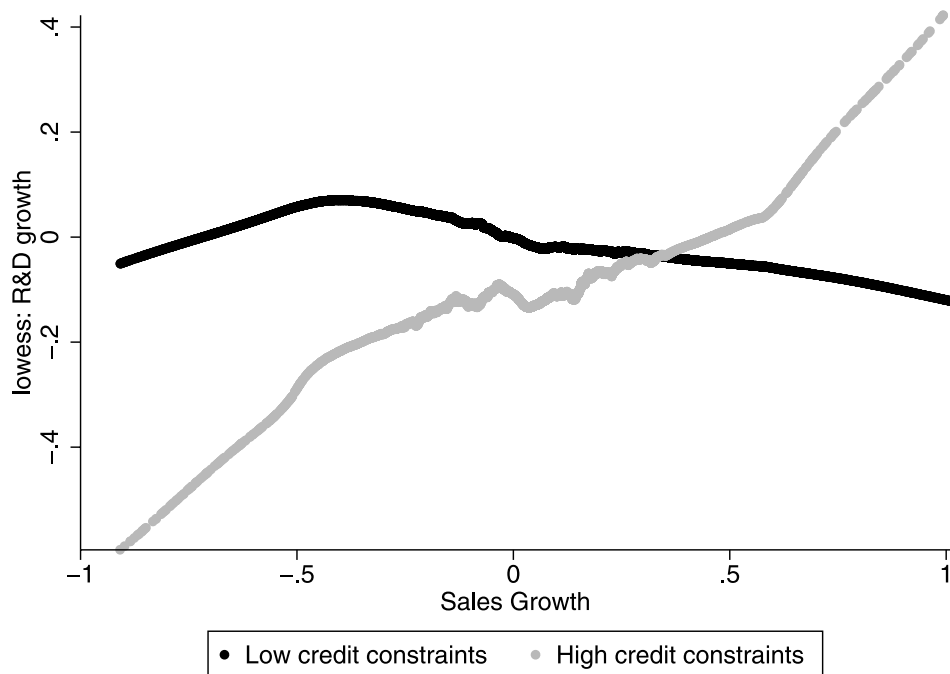


FIGURE 1. Credit constraints and the cyclicality of R&D investment: lowess smoothing.

during downturns: namely, in presence of tighter credit constraints, R&D investment plummets during recessions but it does not increase proportionally during upturns. These results hold even after taking into account potential endogeneity biases.

This has important implications at the macroeconomic level. First, since R&D investment enhances productivity more than short-term investment, our results imply that the cyclical behavior of investment mitigates fluctuations when financial markets are perfect, but amplifies them when credit constraints are sufficiently tight. In other words, tighter credit constraints raises the sensitivity of productivity to exogenous shocks. At the same time, tighter credit constraints reduces the average propensity to engage in R&D investments, by preventing firms from investing in R&D during downturns. It follows that credit constraints contribute to both a lower and more volatile rate of productivity growth.⁴

This paper relates to a broader literature on cycles, innovation and growth. Theoretical papers that take a more Schumpeterian approach to the relation between growth and cycles, include Hall (1991), Gali and Hammour (1992), Caballero and Hammour (1994), Aghion and Saint-Paul (1998), Francois and Lloyd-Ellis (2003), Comin and Gertler (2006), and Barlevy (2004, 2007), however they do not emphasize

4. The working paper version of this work contains suggestive empirical evidence supporting these effects (see Aghion et al. 2008)

the role of credit constraints.⁵ The empirical literature on the subject starts with Ramey and Ramey (1995) who provide cross-country evidence of a negative relationship between volatility and growth. More closely related to the analysis in this paper is AABM. Based on cross-country panel data over the period 1960–2000, AABM show that structural investment (another proxy for growth-enhancing investment) is less countercyclical in countries with lower ratios of credit to GDP, and that the correlation between macroeconomic volatility (measured as in Ramey and Ramey, 1995 by the variance of growth rate) and average growth, is more negative the lower financial development. However, unlike in this paper, the data in AABM do not include R&D investments; moreover, credit constraints are not measured at the firm level. Prior evidence on R&D investments over the cycle is provided by Griliches (1990), Comin and Gertler (2006), and Barlevy (2007), although not in relation to firms' credit constraints.⁶ Using a sample of 1,400 US firms, Brown, Fazzari, and Petersen (2009) find no direct impact of sales changes on R&D while variables that measure access to both internal and external equity finance have a significant effect on firms' R&D expenses; however, these authors do not interact sales shocks and financial constraints. Finally, our results on the asymmetric response of firm-level R&D investment along the business cycle is consistent with recent evidence found at the sectoral level by Ouyang (2011).

The paper is organized as follows. Section 2 presents a simple model to derive our main predictions. Section 3 presents the data and the measurement variables, and in particular our measure of credit constraints. Section 4 develops the key results. Section 5 concludes.

2. Model

The model in this section is a variant of Aghion et al. (2010), but with idiosyncratic sales shocks, and we present it mainly as a guide to our subsequent empirical exercise.

2.1. Basic Environment

Time is discrete, and in each period there is a continuum of two-period lived risk-neutral entrepreneurs who maximize their end-of-life wealth. An entrepreneur born at

5. Our paper is also related to the literature focusing on the effect of uncertainty on R&D investment when there are adjustment costs for changing R&D. As shown by Bloom (2007) or Bloom, Bond, and Van Reenen (2007), the presence of adjustment costs may reduce the responsiveness of R&D investment to demand shocks under uncertainty. If uncertainty is higher during downturns, this may explain the lack of response of R&D during bad times (i.e. the lack of countercyclicality). While this effect is potentially important, it is unlikely to bias our estimates: our proxy for credit constraint is almost uncorrelated with firm-level growth volatility (correlation is 0.02), and we also control for yearly sector-specific volatility through the inclusion of sector $\times$ year dummies.

6. Barlevy (2007) finds no evidence of current cash flows affecting how firms' current R&D investments respond to the business cycle. However, in Barlevy's own estimations, lagged cash flows turn out to significantly affect how current R&D investment reacts to the firm's current position in the business cycle.

date t faces a sales shock a_t at time t and a_{t+1} at time $t + 1$, where $a_t \in \{\underline{a}, \bar{a}\}$, with $1 < \underline{a} < \bar{a}$, and where

$$p = \Pr(a_{t+1} = \bar{a} \mid a_t = \bar{a}) = \Pr(a_{t+1} = \underline{a} \mid a_t = \underline{a})$$

is strictly less than one but greater than $1/2$, so that there is some persistence to a sales shock over time.

At the beginning of her first period, an entrepreneur born at date t is endowed with wealth w , which she can devote to short-run capital investment or to long-run R&D investment. Short-run capital investment k_t yields short-run profit $a_t k_t$ at the end of the first period. Long-term R&D investment z_t yields an innovation value v_{t+1} equal to the expected productivity $E(a_{t+1} \mid a_t)$ in period $(t + 1)$ whenever innovation is successful in the second period. The opportunity to innovate occurs with probability z_t , but innovation is truly successful only if the firm overcomes a liquidity shock (in the form of a cost overrun) which occurs at the end of the first period.

In the absence of credit market imperfections the entrepreneur is always able to overcome the cost overrun, and therefore the probability of innovation is simply equal to z_t in that case. Credit market imperfections however may prevent a firm with short-run profit flow $a_t k_t$ from investing more than $\mu a_t k_t$ in financing the cost overrun, where $\mu \geq 1$ measures the extent to which the firm can borrow using its first period return as collateral. Now, if we assume that the overrun cost c is uniformly distributed between zero and one, then the innovation probability for a credit-constrained firm which invests z_t in R&D will be

$$P_t = z_t \Pr(c \leq \mu a_t k_t) = z_t \min\{1, \mu a_t k_t\}.$$

The timing of moves is as follows. At the beginning of date t the entrepreneur makes her investment decisions. Then at the end of period t short-run profits are realized and also the extent of cost overruns. Then innovation occurs with probability z_t in the following period provided the cost overrun has been financed.

2.2. Profit Maximization and Optimal Investments

Assuming a zero interest rate for notational simplicity, we first consider the case where the entrepreneur born at date t is not credit constrained. Then the entrepreneur will choose k_t and z_t to solve⁷

$$\begin{aligned} \max_{k_t, z_t} \quad & a_t k_t + E(a_{t+1} \mid a_t) z_t \\ \text{subj. to:} \quad & k_t + z_t = w. \end{aligned}$$

7. The linearity of short- and long-run profits in k_t and z_t , together with the assumptions that $1 < \underline{a} < \bar{a}$, and that entrepreneurs are risk-neutral and consume at the end of their lives, implies that in the absence of credit constraints entrepreneurs will invest all their initial wealth either in physical capital or in R&D, depending on the relative rate of returns on these two factors.

This immediately yields

$$z_t = \begin{cases} w & \text{if } E(a_{t+1} | a_t) > \underline{a}, \\ 0 & \text{otherwise,} \end{cases}$$

or equivalently

$$z_t = \begin{cases} w & \text{if } a_t = \underline{a}, \\ 0 & \text{otherwise.} \end{cases}$$

In other words the R&D investment z_t is countercyclical—that is, lower when sales are high with $a_t = \bar{a}$ than when sales are low with $a_t = \underline{a}$. This is the opportunity cost effect already mentioned in the introduction.

Now, consider the case where the entrepreneur is credit-constrained ex post when facing the liquidity shock. Then she will choose k_t and z_t to solve

$$\begin{aligned} \max_{k, z} \quad & a_t k + E(a_{t+1} | a_t) z \min\{1, \mu a_t k_t\} \\ \text{subj. to:} \quad & k_t + z_t = w. \end{aligned}$$

Restricting attention to the case where $\mu a_t k_t < 1$ in equilibrium for both values of a_t , we get that whenever it is interior the optimal solution is simply⁸

$$z_t(a_t) = \frac{1}{2} \left(w - \frac{1}{\mu E(a_{t+1} | a_t)} \right).$$

In particular, this optimal z_t under binding credit constraints is procyclical—that is, with $z_t(a_t = \bar{a}) > z_t(a_t = \underline{a})$, since

$$E(a_{t+1} | a_t = \bar{a}) = p\bar{a} + (1 - p)\underline{a} > E(a_{t+1} | a_t = \underline{a}) = p\underline{a} + (1 - p)\bar{a}.$$

Moreover, the lower μ , namely the tighter the credit constraint, the larger the difference $[z_t(a_t = \bar{a}) - z_t(a_t = \underline{a})]$ —namely, the more procyclical is the optimal R&D investment.

Finally, note that

$$\frac{\partial z_t}{\partial \mu} = \left(\frac{1}{\mu^2} \right) \left(\frac{1}{E(a_{t+1} | a_t)} \right)$$

will be lower when $a_t = \bar{a}$ than when $a_t = \underline{a}$. In other words, a tightening of the credit constraint—that is, a reduction in μ —will have a smaller effect on equilibrium R&D in a boom than in a slump.

8. The solution is interior whenever $w > (\mu E(a_{t+1} | a_t))^{-1}$, which we implicitly assume here.

2.3. Main Theoretical Predictions

The main predictions that emerge from our analysis in this section can be summarized as follows:

1. The more credit constrained the firm is—that is, the lower μ —the more procyclical its R&D investment, in the sense that this investment reacts more positively to the firm's current sales.
2. Tighter credit constraints interact with sales in an asymmetric fashion over the business cycle: a tightening of credit-constraints reduces the firm's R&D investment more in a downturn than in an upturn.

In the remaining part of the paper we take these predictions to French firm-level panel data.

3. Data

Our empirical analysis merges two French firm-level datasets: FiBen, which contains among others balance sheet information, and the payment incident dataset. We first present the FiBen database, and then the payment incident variable, which we use as a proxy for (firm-level) credit constraints.

3.1. The FiBen Database

Our core data comes from FiBen, a large French-firm-level database constructed by the Banque de France. FiBen is based on fiscal documents, including balance sheets and profit and loss statements, and thus contains detailed information on both flow and stock accounting variables. A subsample of FiBen, called *Centrale des Bilans*, is available for a lower number of firms and includes additional information directly collected by the Banque de France. These additional data allow us to perform additional consistency and accuracy tests.

The FiBen database includes all French firms with sales at least equal to €75,000 per year or with credit outstanding of at least €38,000; annual accounting data are available for about 200,000 firms. In 2004, FiBen was covering 80% of the firms with 20 to 500 employees, and 98% of those employing more than 500 employees.⁹

We restrict our sample to firms that have a positive R&D investment during at least one year. Our sample is unbalanced and includes about 13,000 firms over the period 1994–2004. On average each firm appears in seven (not necessarily consecutive) years. We will use only nominal variables but we also include sector–year fixed effects in our regressions, which act as sectoral deflators.

9. More than 50% of the firms in FiBen have less than 20 employees. However, these firms are under-represented in FiBen since their sales rarely exceed the required thresholds.

3.2. *R&D Variable*

Restricting attention to variables for which FiBEn data are available, we choose to concentrate on R&D investment rather than on R&D expenditures as a proxy for long-term, productivity-enhancing investment. R&D investments correspond to the fraction of R&D expenditures which firms are allowed to capitalize. A main reason for us to concentrate on investment rather than on expenditures, is that data on R&D expenditures are only available for (a subset of) large firms in the sample, but the interaction between business cycles and credit constraints is likely to be stronger in smaller firms. Nonetheless, we note that, within the subsample of large firms, R&D investment is strongly correlated with R&D expenditure: a 1% increase in investment is associated with a 0.97% rise in R&D expenditure.¹⁰

3.3. *Description of the Payment Incident Variable*

Although direct information on credit constraints is not available, we construct an indirect measure of credit constraints as follows. Since its introduction in 1992, all French banks have a legal obligation to report within four business days to the “Système Interbancaire de Télécompensation” anytime a firm fails to pay its trade creditors. These defaults on trade credits are called payment incidents (PIs). The Banque de France aggregates this information and makes it available to all commercial banks through a weekly paper or an electronic report automatically sent to all bank agencies. Also, since 1992, through a specific commercial network system, banks can immediately access reports covering the last twelve months; access through the Internet has become available since 2000. The complete longitudinal dataset is available for research only at the Banque de France.

Banks are then supposed to adapt their credit supply to this information, in particular they typically reduce future lending to firms that failed to repay their trade creditors.¹¹ Our indicator for credit constraints ($PI(t)$) is a binary variable equal to 1 when the firm has experienced at least one payment incident during the year, and to zero otherwise.¹² This variable is easy to interpret and it is weakly correlated with all our other variables.¹³ Note that each year, about 7% of firms experience at least one payment incident, and about one-third of firms in our sample has experienced at least

10. We also find that sectoral R&D investment is the lowest in agriculture and the highest in services, in particular business software. Also, average R&D investment growth is positively and significantly correlated with average TFP growth at the firm level in our sample.

11. The payment incidents database is constructed by the Banque de France with the sole purpose of providing information to banks or other credit institutions about their customers. It is therefore natural to expect banks to adapt their credit supply to this information.

12. Alternatively, we also use the number of payment incidents and the value of the unpaid trade credits. We find that payment incidents have nearly the same effects on R&D over the business cycle no matter the number or magnitude of incidents.

13. For instance, its correlation with sales growth is -0.04 , and its correlation with the log of R&D investment is 0.03 .

TABLE 1. Descriptive statistics, whole sample.

Variable	Obs.	Firms	Mean	SD	Q1	Median	Q3
FiBEn							
Whole sample							
No. employees	83,803	13,634	91.75	283.56	15	31	65
$\Delta \log$ Sales	83,803	13,634	0.04	0.19	-0.05	0.04	0.13
R&D investment/Value added	83,536	13,597	0.01	0.03	0	0	0
R&D investment ^{de}	83,803	13,634	29.7	322.96	0	0	0
Payment incident (<i>PI</i>) ^a	83,803	13,634	0.07	0.26	0	0	0
TFP ^d	74,159	11,652	23.43	14.94	14.22	20.51	29.02
Credit-constrained firms^a							
No. employees	30,572	4,844	107.21	325.34	16	32	69
$\Delta \log$ Sales	30,572	4,844	0.04	0.19	-0.06	0.04	0.13
R&D investment/Value added	30,480	4,830	0.01	0.04	0	0	0
R&D investment ^d	30,572	4,844	27.0	189.26	0	0	0
Payment incident (<i>PI</i>) ^a	30,572	4,844	0.2	0.4	0	0	0
TFP ^d	27,033	4,115	22.21	11.92	14.51	20.2	27.68
Non-credit-constrained firms^b							
No. employees	53,231	8,790	82.87	256.08	15	30	63
$\Delta \log$ Sales	53,231	8,790	0.04	0.19	-0.05	0.04	0.13
R&D investment/Value added	53,056	8,767	0.01	0.03	0	0	0
R&D investment ^d	53,231	8,790	31.2	378.9	0	0	0
TFP ^d	47,126	7,537	24.13	16.38	14.04	20.7	29.96
Payment incident table							
No. employees	54,266	11,911	91.65	266.56	17.00	33.00	69.00
New bank loans/VA	54,253	11,908	0.03	1.37	0.00	0.00	0.01
Long-term/Total loans	51,596	11,622	0.29	0.27	0.00	0.24	0.57
Collateral ^c	54,266	11,911	11,375	1.0e+05	645	1611	4593
Bank debt/Total financing	53,732	11,843	0.18	0.15	0.05	0.16	0.28

Notes: ^a At least one payment incident during the period; ^b No payment incident during the period; ^c Sum of fixed and tangible assets. ^d Thousands of euros. ^e Positive R&D investment for 23% of the total number of observations. Source: Authors' computations from Fiben/Centrale des Bilans, Banque de France.

one payment incident over the whole period. All sectors are concerned by payment incidents, especially manufacturing motor vehicles which include small and medium subcontractors facing high-magnitude business cycles.

Table 1 provides summary statistics for our main variables including sales, R&D investment, and our measure of credit constraint. This measure, which is referred to as "payment incident", will be described and analyzed in detail in what follows. Our final sample includes a large number of small and medium size firms¹⁴, that are particularly prone to be hit by credit constraints. Moreover, Table 1 shows that credit-constrained firms (here defined as those firms that have experienced at least one payment incident during the period) display a lower R&D investment and total factor productivity.¹⁵ This is consistent with our theoretical predictions: if credit constraints

14. The median firm size in our sample is of around 30 employees per firm.

15. Our TFP measure includes three factors of production—capital, labor, and intermediate inputs. Capital is deflated using a gross fixed-asset deflator from the OECD Economic Outlook database. For value added, we use a sectoral deflator from the EU-KLEM database.

are binding, productivity-enhancing investments become less countercyclical (or even procyclical). Our empirical analysis in the next section will confirm these stylized facts.

Before we proceed with the empirical analysis, it is worth pointing out that experiencing a payment incident does not mean that the firm is necessarily in financial distress or close to bankruptcy. Around one-third of all firms in our dataset experience at least once a payment incident (see Table 1), suggesting that payment incidents hit a much larger set of firms than those about to disappear. As a robustness check, we have restricted our analysis to firms remaining in our sample over the entire period, or to firms which are still in the sample four years after the payment incident—that is, firms for which having a payment incident did not mean being close to bankruptcy. Such restrictions, available upon request, left our results unchanged.

3.4. *Payment Incidents as a Generator of Credit Constraints*

In this section we investigate the effect for a firm of experiencing a PI on its future bank loans. More precisely, we study the impact of having experienced at least one PI over the two previous years ($t - 1$ and $t - 2$) both on the probability to contract a new bank loan, and on the amount of this loan in year t . We thus estimate the following equation:

$$\log(BkL_{i,t} + 1) = \alpha_1 PI_{i,t-1} + \alpha_2 PI_{i,t-2} + \beta \mathbf{X}_{i,t-1} + \mu_{kt} + \rho_i + \varepsilon_{i,t}, \quad (1)$$

where $BkL_{i,t} \geq 0$ represents the amount of new bank loans contracted by firm i during year t , $PI_{i,t-j}$ is a binary variable equal to 1 whenever firm i had a PI during year $t - j$, and $\mathbf{X}_{i,t-1}$ is a vector of controls. While past PIs are expected to have a negative effect on bank supply, we have to control for factors affecting both credit supply and demand, as well as for variables which may be correlated to both $BkL_{i,t}$ and $PI_{i,t-1}$.

On the supply side, we include controls for firm size (number of employees), for the firm's cash-flow, for collateral and the firm's dependence upon bank finance (banking debt over total debt).¹⁶ All these variables but the last one are in logarithms, and all are lagged by one period. We expect the supply of bank loans to be higher for firms with higher cash flow and collateral. Size is expected to have a nonlinear effect—that is, a lower positive effect on larger firms. Finally, we expect the estimated coefficients on the PI variable to be negative—banks are supposed to reduce their credit supply to firms that failed to repay their trade creditors.

Other variables are likely to affect the demand for bank loans and both the probability of experiencing a payment incident and the level of credit. We thus include a full set of sector $\times$ year dummies (μ_{kt}), to account for macroeconomic or sectoral variations in credit supply and demand. We also control for sales and productivity growth in the current and last two periods, as well as for R&D investment. Finally, we include firm fixed effects (ρ_i).

16. A more detailed description of the computation of these different variables is provided in Table A.1.

In addition to this main regression exercise, we also separately assess the impact of having experienced a payment incident in the past, respectively on the access to new bank loans and on the amount of new bank loans, using a Tobit estimation. Finally, we replace the dependent variable “new bank loans” by the share of long-term loans over total loans. The idea is that credit-constrained firms should rely relatively more on short-term loans as banks are more reluctant to extend long-term loans to such firms. We thus expect the coefficient on *PI* to be negative in this latter estimation.

The results are shown in Table 2. Having experienced a PI during the previous year has a negative and significant impact on the amount of new bank loans. In the last two columns we decompose this overall effect into a marginal effect on the probability to contract a new loan and the effect on the size of the loan. Having experienced a PI has more negative impact on the size of future loans (column (n)) than on the probability of contracting a new loan (column (m)). We also find that having experienced a PI two years before does not have any impact on current credit supply. One potential explanation for this latter finding is that the electronic service provided by the Banque de France gives commercial banks access to only the past year *PI*.

These results are quite stable across estimations, and also robust to the inclusion of the different control variables. Coefficients on the main right-hand side variables also have the expected signs: a larger cash flow, size and collateral are all positively correlated with the amount of bank credit supply. Results are also qualitatively unchanged when controlling for current past sales growth (columns (d) to (f)), R&D investment (column (e)), TFP growth (column (f)) and the lag of the dependent variable (columns (g) and (h)). Note that banks' access to PI data via the internet since 2000 does not seem to have modified the correlation between *PI* and credit supply (columns (i) and (j)). Finally, columns (k) and (l) show a negative correlation between *PI* and the share of long-term loans in total loans—which in turn has implications for the relationship between credit constraints and long-run R&D investment.

These findings are consistent with the idea of a significant impact of PIs on credit supply. We shall build on these results in our main analysis, in which we use the binary variable equal to 1 whenever the firm has experienced at least one *PI* in year $t - 1$, as our indicator for credit constraint in year t .

As we explain in more detail in the next section, this measure of credit constraint is not immune from potential endogeneity problems. In particular, both R&D investment and the fact of having experienced a PI may result from the existence of omitted variables. For example, the firm may decide that a given activity is no longer worth pursuing, and as a result reduce both, its R&D investment and also its diligence vis-à-vis trade creditors in that activity. To deal with the endogeneity problem and further confirm the relevance of PIs as a generator of credit constraints, we use the Rajan and Zingales (1998) industry-level measure of financial external dependence.¹⁷ More precisely, we run our main estimations on two subsamples of sectors, respectively with US counterparts that display higher and lower levels of external financial dependence

17. See Rajan and Zingales (1998). The RZ indicator measures the extent to which the corresponding sector in the US is more or less dependent upon external finance.

TABLE 2. Payment incidents as a generator of credit constraints.

Dep. var. :	WITHIN										WITHIN		TOBIT	
	log (New bank loans)										Long-term loans/ Total loans		log (New bank loans) $P(X > 0)$ $E(X X > 0)$	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i) Before 2000	(j) After 2000	(k)	(l)	(m)	(n)
$PI(t-1)$	-0.262*** (0.038)	-0.256*** (0.040)	-0.234*** (0.040)	-0.224*** (0.042)	-0.202*** (0.043)	-0.195*** (0.044)	-0.224*** (0.047)	-0.180*** (0.050)	-0.228** (0.103)	-0.252*** (0.053)	-0.021*** (0.003)	-0.021*** (0.003)	-0.043*** (0.007)	-0.180*** (0.029)
$PI(t-2)$		-0.071* (0.040)	-0.067* (0.040)	-0.055 (0.043)	-0.026 (0.043)	-0.022 (0.044)	-0.018 (0.045)	0.025 (0.049)	-0.197* (0.105)	-0.04 (0.050)	-0.014*** (0.003)	-0.014*** (0.003)	-0.005 (0.007)	-0.019 (0.030)
$\log \text{Cash-flow}(t-1)$			0.394*** (0.076)	0.398*** (0.087)	0.385*** (0.085)	0.353*** (0.120)	0.326*** (0.089)	0.377*** (0.138)	0.464*** (0.172)	0.253*** (0.092)	0.066*** (0.008)	0.098*** (0.013)	0.290*** (0.018)	1.199*** (0.077)
$\log \text{No. employees}$ $(t-1)$		0.114 (0.124)	0.114 (0.124)	0.043 (0.131)	0.000 (0.148)	-0.175 (0.195)	-0.041 (0.150)	-0.418* (0.226)	-0.074 (0.342)	0.092 (0.177)	-0.007 (0.009)	0.002 (0.014)	0.093*** (0.010)	0.386*** (0.034)
$(\log \text{No. employees})^2$ $(t-1)$		-0.025 (0.019)	-0.025 (0.019)	-0.017 (0.020)	-0.011 (0.022)	0.000 (0.027)	0.000 (0.024)	0.03 (0.032)	0.006 (0.049)	-0.022 (0.028)	0.000 (0.001)	-0.003* (0.002)	-0.013*** (0.001)	-0.056*** (0.004)
$\log \text{Collateral}$ $(t-1)$		0.344*** (0.030)	0.344*** (0.030)	0.332*** (0.033)	0.309*** (0.041)	0.461*** (0.046)	0.371*** (0.037)	0.484*** (0.055)	0.315*** (0.076)	0.363*** (0.040)	0.011*** (0.002)	0.023*** (0.004)	0.018*** (0.002)	0.075*** (0.009)
$\text{Bank dep.}(t-1)$		-1.359*** (0.136)	-1.359*** (0.136)	-1.386*** (0.142)	-1.366*** (0.147)	-1.422*** (0.150)	-1.877*** (0.167)	-1.902*** (0.185)	-3.097*** (0.355)	-1.587*** (0.175)	0.262*** (0.011)	0.271*** (0.012)	0.290*** (0.016)	1.202*** (0.069)
$\Delta \log \text{Sales}(t)$				0.034 (0.031)	0.148*** (0.047)	0.212*** (0.057)	0.172*** (0.064)	0.172*** (0.064)			-0.008* (0.004)	-0.008* (0.004)		
$\Delta \log \text{Sales}(t-1)$				0.065* (0.030)	0.117*** (0.037)	0.379*** (0.047)	0.347*** (0.073)	0.347*** (0.073)			0.011*** (0.004)	0.011*** (0.004)		
$\Delta \log \text{Sales}(t-2)$				0.114*** (0.034)	0.153*** (0.044)	0.216*** (0.063)	0.231*** (0.053)	0.231*** (0.053)			0.013*** (0.004)	0.013*** (0.004)		
$\log (\text{R\&D investment})$					0.054*** (0.010)	0.053*** (0.011)	0.054*** (0.012)	0.054*** (0.012)			-0.001 (0.001)	-0.001 (0.001)		
$\Delta \log \text{TFP}(t)$						-0.176*** (0.040)	-0.196*** (0.044)	-0.196*** (0.044)			0.004 (0.003)	0.004 (0.003)		
$\Delta \log \text{TFP}(t-1)$						-0.235*** (0.050)	-0.263*** (0.044)	-0.263*** (0.044)			-0.008** (0.003)	-0.008** (0.003)		
$\Delta \log \text{TFP}(t-2)$						0.036 (0.040)	0.044 (0.045)	0.044 (0.045)			-0.001 (0.003)	-0.001 (0.003)		
$\log (\text{New bank loans}(t-1))$						0.052*** (0.007)	0.052*** (0.007)	0.052*** (0.007)						
Observations	54,266	51,207	50,667	47,578	45,387	44,412	40,846	36,076	14,473	36,194	58,377	50,969	50,667	50,667
Firms	11,911	11,385	11,310	10,664	10,432	10,215	10,148	9,252	8,167	10,124	12,106	10,886	11,310	11,310
Adjusted R ²	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.01	0.04	0.04		

Notes: Robust standard errors into parentheses. All variables are computed from Friben/Centrale des Bilans, Banque de France. PI : Payment incident (0/1). Bank dep.: (Banking debt/Total debt). As the variables "New Bank loans" and "R&D investments" can take the value 0, we use $\log(x+1)$ for these variables in the within specifications. All estimations include sector $\times$ year dummies. Marginal effects computed at means for Tobit estimations.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

(compared to the median). We explain our methodology in more detail in the next section.

4. Credit Constraints and the Cyclicity of R&D Investment

In this section we use the *PI* measure of credit constraints to test our main theoretical predictions. In particular we will show that: (1) R&D investment is procyclical for firms facing tighter credit constraints; (2) this procyclicality effect tends to be asymmetric: it operates only during low sales states. The next section will discuss robustness checks and implications of our results.

4.1. Cyclicity of R&D and Credit Constraints

4.1.1. Specification. We test our first proposition by estimating the following equation:

$$\begin{aligned} \log(RD_{i,t} + 1) = & \sum_{j=0}^2 \beta_j \Delta \log s_{i,t-j} + \theta PI_{i,t-1} \\ & + \sum_{j=0}^2 \gamma_j \Delta \log s_{i,t-j} \times PI_{i,t-1} + \mu_{kt} + v_i + \varepsilon_{it}, \end{aligned} \quad (2)$$

where RD_{it} represents current R&D investment (used as a proxy for long-term, productivity-enhancing investment), $PI_{i,t-1}$ the *PI* dummy (used as an indicator for credit constraints), and Δs_{it} firm i 's sales variation (sales increase) during year t . We control for firm fixed effects and we again include a full set of year $\times$ sector dummies μ_{kt} ; these allow, among others, to capture time variations in sales, and they also act as R&D deflators.

Based on our theoretical analysis, we expect R&D investment to be countercyclical *absent of any credit constraints*; we thus expect $\sum \beta_j < 0$. However, tighter credit constraints should reverse the *cyclicity* of R&D investment—that is, $\sum \gamma_j > 0$. The comparison between $\sum \beta_j$ and $\sum \beta_j + \sum \gamma_j$ provides information on the cyclicity of R&D investment in more versus less credit constrained firms.

4.1.2. Results. Table 3 reports the results on the impact of current and past (one and two period lagged) sales growth variables on current R&D investment.¹⁸ Without introducing the *PI* variable, R&D investment is only weakly countercyclical: only the coefficients on the first lag of the sales growth variable are significant at 10%; however, the sum of all sales growth regression coefficients, $\sum \beta_j$, is negative (−0.11) and significant at the 10% level (column (c)).

18. Including t -period lagged sales growth variables with $t > 2$ in the right-hand side of the estimation equation does not modify the results, and the regression coefficients on these variables are all insignificant.

TABLE 3. Credit constraints and the cyclicity of R&D investment: Basic estimations.

Dep. var.:	log($RD + 1$)					
	(a)	(b)	(c)	(d)	(e)	(f)
$\Delta \log Sales(t)$	-0.032 (0.027)	-0.04 (0.027)	-0.042 (0.028)	-0.062** (0.028)	-0.071** (0.029)	-0.073** (0.029)
$\Delta \log Sales(t - 1)$		-0.049* (0.026)	-0.052* (0.028)		-0.070*** (0.027)	-0.074*** (0.029)
$\Delta \log Sales(t - 2)$			-0.015 (0.027)			-0.033 (0.028)
$PI(t - 1)$				0.001 (0.022)	-0.007 (0.022)	-0.017 (0.021)
$\Delta \log Sales(t) * PI(t - 1)$				0.355*** (0.100)	0.368*** (0.100)	0.371*** (0.100)
$\Delta \log Sales(t - 1) * PI(t - 1)$					0.278*** (0.100)	0.285*** (0.102)
$\Delta \log Sales(t - 2) * PI(t - 1)$						0.229** (0.097)
$\sum \beta_j$			-0.110* (0.033)			-0.180*** (0.064)
$\sum \beta_j + \sum \gamma_j$						0.704*** (0.178)
No. obs.				83,803		
No. groups				13,634		
Estimation				Within		

Notes: Robust standard errors into parentheses. All estimations include sector $\times$ year dummies.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

Introducing the PI measure of credit constraints and its interaction with sales growth greatly modifies the results (columns (d) to (f)): the coefficients on sales growth variables become more significantly negative, and their interactions with the PI variable are positive and significant. The overall cyclicity for less constrained firms (column (f)) is equal to -0.19 and it is strongly significant. For more-credit-constrained firms, it is equal to 0.70 and is also significant at the 1% level, meaning that credit constraints actually make R&D investment become *procyclical*. Note that our results are robust to the use of a Tobit or of a Poisson Pseudo-maximum likelihood estimator which account for the high number of zeros in our R&D variable. Finally, estimating the cyclicity of the *probability* of undertaking R&D investment using a fixed effects Logit estimation leads to similar results.¹⁹

4.1.3. Robustness Check: Simultaneity. The estimated coefficients may be biased as current sales and R&D investment may be co-determined. A first way to deal with this endogeneity problem is to use an instrumental variable (IV) methodology where the instruments consist of an appropriate set of lagged values of the regressors. However, such instruments are typically weak, and tests of over-identifying restrictions do not always support them. To deal with the potential simultaneity bias, we choose to perform

19. Results are available upon request.

a two-stage estimation²⁰ using two additional instrumental variables for current sales growth. We make use of destination-specific information on firms' exported values from the French customs to construct two different indicators,²¹ respectively reflecting the levels of real exchange rate and foreign demand faced by exporters. More precisely, we construct the following indicators:

$$RER_{it} = \left(\sum_{j=1}^N rer_{jt} \times \alpha_{ij0} \right) \times \frac{X_{i,t-1}}{S_{i,t-1}}, \quad (3)$$

$$GDP_{it} = \left(\sum_{j=1}^N gdp_{jt} \times \alpha_{ij0} \right) \times \frac{X_{i,t-1}}{S_{i,t-1}}, \quad (4)$$

where rer_{jt} is the real bilateral exchange rate between France and country j , gdp_{jt} is the GDP of country j , α_{ij0} is the share of firm i 's total export to country j at the beginning of the period, X_{it} and S_{it} are firm i 's total exports and total sales during year t . For each indicator, the first term represents the average real exchange rate and the average foreign demand faced by firm i , weighted by the share of each destination in firm i 's total exports. The second term measures firm i 's degree of openness in period $t - 1$. Both, a real depreciation (decrease in rer) and an increase in foreign demand (increase in gdp) have a positive impact on exports, and thereby a positive effect on current sales, especially when the firm is more export oriented. These instruments are firm-year specific, but the exchange rate and GDP levels are determined at the macro level, so that they are arguably exogenous to firms' decisions, and in particular to its level of R&D investment. In addition to the differentiated lag of our regressors, we use current and lagged values of these two indicators as instruments.²² All sales and PI variables are instrumented for, and we check the validity of the corresponding instruments using the Hansen test for over-identifying restrictions.

Columns (a) and (b) of Table 4 present the results of the two-stage estimations. Column (a) uses standard instruments—that is, two differentiated lags of the regressors. In addition to the differentiated lag of our regressors, we use in column (b) the current value and two lags of RER_{it} and GDP_{it} as external instruments; these are more clearly validated by the test of over-identifying restrictions.²³ In line with the previous results, R&D investment turns out to be clearly procyclical in presence of tighter credit constraints.

20. We also ran dynamic GMM estimations. The results were not altered, neither qualitatively nor quantitatively, but instruments were generally rejected by the Sargan test of over-identifying restrictions. This is in line with a number of papers (including Hall, Mairesse, and Mulkay 1999) which emphasize the weakness of GMM instruments in this kind of firm-level estimations.

21. Customs data contain, for each exporter, the value of exports by year and destination. For a description of these data, see for instance Eaton, Kortum, and Kramarz (2004).

22. We also tried to use more lags, or older lags of these instruments in the 2SLS specification. Although this results in an important loss of observations, the results remained qualitatively unchanged.

23. The sample size is reduced due to data availability: we only have destination-specific export information from 1996 to 2004 (due to the use of lagged values of the instrumental variables, the time interval used in the estimation is 1998–2004). Running the within estimations on this subsample does not modify the results.

TABLE 4. Credit constraints and the cyclicalty of R&D investment: robustness.

Dep. var.: Subsample	$\log(RD + 1)$									
	Asset tangibility					Financial dependence				
	(a)	(b)	Low (c)	High (d)	Low (e)	High (f)	Low (g)	High (h)	Low (i)	High (j)
$\Delta \log Sales(t)$	-0.005 (0.041)	-0.028 (0.060)	-0.305*** (0.069)	0.080 (0.067)	-0.288** (0.144)	0.139 (0.132)	0.007 (0.060)	-0.236*** (0.062)	0.075 (0.130)	-0.172 (0.124)
$\Delta \log Sales(t - 1)$	-0.069* (0.040)	-0.089 (0.063)	-0.276*** (0.066)	-0.032 (0.065)	-0.267* (0.149)	-0.039 (0.131)	-0.059 (0.056)	0.212*** (0.060)	0.007 (0.127)	-0.140 (0.128)
$\Delta \log Sales(t - 2)$	-0.036 (0.039)	-0.077 (0.059)	-0.176*** (0.065)	-0.069 (0.063)	-0.135 (0.140)	-0.090 (0.126)	-0.050 (0.052)	-0.172*** (0.060)	-0.087 (0.117)	-0.088 (0.123)
$PI(t - 1)$	-0.021 (0.028)	-0.088* (0.047)	-0.033 (0.055)	-0.035 (0.046)	-0.211** (0.102)	0.000 (0.103)	-0.006 (0.038)	-0.016 (0.049)	0.010 (0.083)	-0.148 (0.098)
$\Delta \log Sales(t) * PI(t - 1)$	0.118 (0.138)	0.419** (0.202)	0.532** (0.228)	0.297 (0.224)	0.897** (0.412)	0.242 (0.487)	0.387* (0.197)	0.500** (0.200)	0.464 (0.431)	0.724** (0.386)
$\Delta \log Sales(t - 1) * PI(t - 1)$	0.234** (0.142)	0.376* (0.218)	0.525** (0.230)	-0.139 (0.221)	1.006** (0.494)	-0.569 (0.538)	0.189 (0.198)	0.316 (0.201)	-0.492 (0.447)	0.613 (0.437)
$\Delta \log Sales(t - 2) * PI(t - 1)$	0.194 (0.139)	0.310 (0.202)	0.683*** (0.236)	0.177 (0.213)	0.535 (0.490)	0.132 (0.538)	0.102 (0.174)	0.604*** (0.209)	0.033 (0.393)	0.498 (0.470)
$\sum \beta_j$	-0.110 (0.092)	-0.194 (0.152)	-0.757*** (0.138)	-0.021 (0.128)	-0.690* (0.371)	0.010 (0.323)	-0.102 (0.114)	-0.619*** (0.125)	-0.005 (0.309)	-0.401 (0.323)
$\sum \beta_j + \sum \gamma_j$	0.437* (0.254)	0.910** (0.416)	0.984** (0.396)	0.315 (0.359)	1.748* (0.931)	-0.185 (1.004)	0.575* (0.331)	0.801*** (0.341)	0.001 (0.832)	1.434* (0.851)
Obs.	54,085	33,763	16,465	17,481	7,235	7,919	21,874	20,257	9,328	8,859
Firms	9,267	7,187	2,771	2,789	1,526	1,633	3,426	3,351	1,948	1,854
Estimation (1)	FE-2SLS					FE-2SLS				
Overid. (Hansen) stat.	5.734	7.393			9.284	11.52			13.63	9.05
p-value	0.33	0.83			0.678	0.485			0.325	0.699

Notes: Robust standard errors into parentheses. All estimations include sector $\times$ year dummies. Rajan and Zingales (1998) data for sectoral financial dependence. Braun (2003) data for sectoral asset tangibility. (1) Standard instruments, i.e. two differentiated lags of regressors used in column (a). See equations (3) and (4) for the detailed description of the instruments used in the other IV estimations.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

4.1.4. Robustness Check: Omitted Variable. As already mentioned in Section 3, another source of endogeneity lies in the possibility that both a firm's R&D investment and whether it is subject to a PI hinge on some omitted variable. The omitted variables have to be firm-year specific (otherwise, they are captured by year or firm fixed effects), and to co-determine PI in year $t - 1$ and the R&D investment in year t , without affecting the R&D at $t - 1$ in the same way as it affects the R&D at t (since the inclusion of a lagged term of the dependant variable does not modify the results).

Although it is instrumented in the IV equation, the PI variable does not satisfy the rank condition, that is, our external instruments do not predict $PI(t - 1)$ (Table A.2). To deal with this problem, we use the sectoral financial dependence indicator of Rajan and Zingales (1998): we run the last set of estimations on two different subsamples, respectively consisting of sectors with US counterparts that are more and less financially dependent than the median. The intuition behind this exercise is twofold. First, there is a priori no reason for this endogeneity bias to be differently distributed across sectors with different levels of external dependence, that is, for the omitted variable to affect $PI(t - 1)$ and the R&D investment in year t (with the above-mentioned restrictions) only in sectors that are more dependent upon external finance. Second, the previous results should be magnified in more financially dependent sectors. Hence, getting more significant results on the financially dependent subsample would suggest both that the endogeneity bias is weak and that PIs indeed generate firm-level credit constraints. Importantly, the probability of experiencing a PI is not correlated with the sectoral degree of financial external dependence: this probability is around 9% in both subsamples, and the correlation between these two variables is close to zero and statistically insignificant.

We repeat this exercise but using the Braun (2003) index of asset tangibility instead of the measure of external dependence. Firms operating in industries with US counterparts displaying higher levels of asset tangibility²⁴ are expected to face lower credit constraints, everything else being equal. The interaction between sales and PI is thus expected to be more significant for firms operating in industries characterized by a lower level of asset tangibility.

Results are provided in columns (c)–(j) of Table 4. Note that the sample is again reduced because data on financial dependence and asset tangibility are only available for manufacturing sectors. R&D investment is procyclical in presence of tighter credit constraints only in sectors characterized by a low degree of asset tangibility (columns (c) and (e)) or more dependent upon external finance (columns (h) and (j)). Estimated coefficients are in general insignificant in the other subsamples (column (d), (f), (i)). The interaction with current sales growth is significant at the 10% level in column (g), but becomes insignificant in the 2SLS estimation of column (i). This in turn suggests a causal effect of credit constraints on the procyclicality of R&D investments.

24. Tangible assets include net property, plant, and equipment. For more information on the computations of asset tangibility and external dependence indexes, see Braun (2003) and Rajan and Zingales (1998).

4.2. Proposition 2: Asymmetry Between Positive And Negative Shocks

4.2.1. Specification. The interaction terms in the previous tables need to be interpreted with caution: their positive signs is equally consistent with the fact that credit constraints prevent firms from increasing their R&D investment in downturns, or that firms increase more their investment during upturns periods when they are financially constrained.

In this section, we disentangle the up- and downturn effects and show that the impact of credit constraints on R&D investment depends upon the firm's current location within its business cycle. As argued in the theoretical section, one expects this effect to be stronger during downturns as credit constraints are more likely to be binding in that case. To test this prediction, we decompose the sales growth variable in two components: a sales growth during downturns variable (when sales growth is above its mean value) and a sales growth variable during upturns (when sales growth is below its mean value, computed by firm).²⁵

Our prediction is then that the interaction terms between sales growth in downturns and PIs that should be the most positive and significant. More formally, we estimate the equation:

$$\begin{aligned} \log(RD_{i,t} + 1) = & \sum_{j=0}^2 (\alpha_j \Delta s_{i,t-j}^H + \gamma_j \Delta s_{i,t-j}^L) + \alpha_4 PI_{i,t-1} \\ & + \sum_{j=0}^2 (\theta_j \Delta s_{i,t-j}^H \times PI_{i,t-1} + \lambda_j \Delta s_{i,t-j}^L \times PI_{i,t-1}) + \mu_{kt} + v_i + \varepsilon_{it}, \end{aligned} \quad (5)$$

where $\Delta s_{i,t}^H$ equals sales variations if the firm is in an upturn, and to 0 otherwise; $\Delta s_{i,t}^L$ equals sales variations if the firm is in a downturn, 0 otherwise. We then repeat the same estimation exercise, but with $\Delta s_{i,t}^H$ being taken equal to one if the firm's sales growth is above the third quartile (computed by sector) of this variable and is equal to zero otherwise; and with $\Delta s_{i,t}^L$ being equal to one if the firm's sales growth is below its first quartile, and to zero otherwise.²⁶ Our conjecture is that credit constraints should play a more important role during downturns: $\sum \lambda_j > \sum \gamma_j$.

4.2.2. Results. The results are shown in Table 5. The estimations in columns (a)–(c) and (e)–(g) suggest that during upturns R&D investment is more countercyclical in less-credit-constrained firms, but that it appears to be acyclical in more-credit-constrained

25. We thus implicitly assume that a large increase (resp. a decrease) in sales indicates that the firm experiences an upturn (resp. a downturn). However, one may object that even firms in the low (resp. high) part of their business cycle may experience large negative (resp. positive) shocks. To respond to this objection, we have checked the robustness of our results to controlling for the initial position of the firm within its cycle in period $t - 1$. The regression results (available upon request to the authors) remain unchanged.

26. We also performed some robustness checks to alternative decompositions, based on quartiles computed by year, of sector–year, or decomposition between positive and negative sales growth. The results were qualitatively unchanged.

TABLE 5. Credit constraints and the cyclicity of R&D investment, asymmetry (1).

Dep. var.:	$\log(RD + 1)$							
	Decomposition by firm (1)				Decomposition by sector (2)			
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
High $\Delta \log Sales(t)$	-0.049 (0.041)	-0.049 (0.041)	-0.063 (0.042)	-0.011 (0.008)	-0.008 (0.041)	-0.008 (0.041)	-0.023 (0.042)	-0.007 (0.008)
Low $\Delta \log Sales(t)$	-0.027 (0.050)	-0.026 (0.050)	-0.081 (0.052)	-0.005 (0.010)	-0.064 (0.052)	-0.063 (0.052)	-0.118** (0.054)	-0.011 (0.011)
High $\Delta \log Sales(t-1)$	-0.110*** (0.039)	-0.109*** (0.039)	-0.137*** (0.040)	-0.014** (0.008)	-0.105*** (0.039)	-0.105*** (0.039)	-0.126*** (0.040)	-0.013* (0.008)
Low $\Delta \log Sales(t-1)$	0.04 (0.050)	0.04 (0.050)	0.027 (0.052)	-0.002 (0.010)	0.023 (0.052)	0.023 (0.052)	0.004 (0.054)	-0.003 (0.012)
High $\Delta \log Sales(t-2)$	-0.062* (0.038)	-0.062* (0.038)	-0.068* (0.039)	-0.016** (0.008)	-0.036 (0.038)	-0.036 (0.038)	-0.042 (0.039)	-0.015** (0.008)
Low $\Delta \log Sales(t-2)$	0.065 (0.049)	0.065 (0.049)	0.027 (0.051)	0.019** (0.010)	0.03 (0.051)	0.03 (0.051)	-0.008 (0.052)	0.015 (0.011)
$PI(t-1)$		0.006 (0.021)	0.007 (0.031)	0.001 (0.006)		0.007 (0.021)	0.030 (0.029)	0.005 (0.005)
High $\Delta \log Sales(t) * PI(t-1)$			0.244 (0.170)	-0.010 (0.028)			0.273 (0.168)	-0.017 (0.028)
Low $\Delta \log Sales(t) * PI(t-1)$			0.492*** (0.165)	0.093*** (0.036)			0.480*** (0.170)	0.113*** (0.037)
High $\Delta \log Sales(t-1) * PI(t-1)$			0.125 (0.175)	0.051 (0.034)			0.301* (0.158)	0.026 (0.030)
Low $\Delta \log Sales(t-1) * PI(t-1)$			0.394** (0.158)	0.032 (0.031)			0.2 (0.181)	0.067** (0.034)
High $\Delta \log Sales(t-2) * PI(t-1)$			0.074 (0.136)	0.029 (0.030)			0.078 (0.134)	0.030 (0.028)
Low $\Delta \log Sales(t-2) * PI(t-1)$			0.458** (0.182)	0.006 (0.035)			0.463** (0.188)	0.007 (0.036)

TABLE 5. Continued.

Dep. var.:	$\log(RD + 1)$							
	Decomposition by firm (1)				Decomposition by sector (2)			
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
$\sum \alpha_j$	-0.220*** (0.075)	-0.220*** (0.076)	-0.268*** (0.077)	-0.042** (0.020)	-0.150** (0.075)	-0.149** (0.075)	-0.191** (0.077)	-0.036* (0.020)
$\sum \gamma_j$	0.078 (0.096)	0.079 (0.096)	-0.027 (0.098)	0.011 (0.025)	-0.011 (0.099)	-0.010 (0.099)	-0.121 (0.101)	0.000 (0.027)
$\sum \alpha_j + \sum \theta_j$			0.176 (0.291)	0.028 (0.059)			0.461* (0.242)	0.003 (0.052)
$\sum \gamma_j + \sum \lambda_j$			1.318*** (0.313)	0.144*** (0.065)			1.021*** (0.277)	0.189*** (0.059)
No. obs.		83,803		32,253		83,803		32,223
No. firms		13,634		6,962		13,634		6,961
Estimation		WITHIN		FE-2SLS		WITHIN		FE-2SLS
Over-id. (Hansen) stat.				19.8				19.55
p-value				0.707				0.722

Notes: (1) Decomposition by firm: above (high) and below (low) firm's mean sales' variation; (2) Decomposition by sector: firm above the third quartile of its sector's sales variation (high) or below the first quartile (low). Robust standard errors into parentheses. All estimations include sector $\times$ year dummies.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

TABLE 6. Credit constraints and the cyclicalities of R&D investment, weighted estimations.

Dep. var.:	$\log(RD + 1)$			
	(a)	(b)	(c)	(d)
$\Delta \log Sales(t)$	-0.026 (0.017)	-0.060*** (0.017)	-0.042*** (0.010)	-0.071*** (0.010)
$\Delta \log Sales(t - 1)$	-0.054*** (0.016)	-0.075*** (0.016)	-0.060*** (0.010)	-0.082*** (0.010)
$\Delta \log Sales(t - 2)$	-0.027* (0.015)	-0.046*** (0.016)	-0.022** (0.009)	-0.041*** (0.010)
$PI(t - 1)$		-0.040*** (0.012)		-0.030*** (0.008)
$\Delta \log Sales(t) * PI(t - 1)$		0.390*** (0.062)		0.361*** (0.038)
$\Delta \log Sales(t - 1) * PI(t-1)$		0.264*** (0.058)		0.294*** (0.037)
$\Delta \log Sales(t - 2) * PI(t-1)$		0.245*** (0.059)		0.248*** (0.036)
No. obs.		83,803		
No. groups		13,634		
Estimation		Within		
Weighted by	No. workers		Value added	

Notes: Panel, weighted least squares. Robust standard errors into parentheses. All estimations include sector $\times$ year dummies.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

firms. In downturns, however, R&D investment becomes clearly procyclical when the firm is more credit constrained: the interaction terms between sales variation and PI are significant at the 1% level for lower shocks. This is consistent with the view that firms escape their credit constraints when moving upward in their business cycle. Finally, the use of the IV methodology confirms that the interacted term between PIs and sales variations is only significant for low sales shocks (columns (d) and (h)).

4.3. From Micro to Macro Effects of Credit Constraints

That we obtain significant correlations at the individual firm level does not automatically imply that innovation over the cycle should be also affected by credit constraints at the macro level. Indeed, one could argue that the bulk of R&D investment is concentrated on a small number of large firms, which are less likely to be hit by credit constraints. Moreover, given that our previous estimations treat each firm during a particular year as one observation, regardless of firm size, our results may simply suggest that, although significant at the individual level, the effect we find leaves overall R&D roughly unaffected. In this section, we check the robustness of our results to variations in firm size, which we measure by the firm's value added or by the number of employees.

As shown in Table 6, even when weighting the estimations by firm size, the interaction terms between PI and sales variations remain highly significant. Only

the payment incidents variable becomes negative and significant. Thus, our main conclusion about R&D investment being more procyclical in more credit constrained environments, should carry over from firm-level to the aggregate economy.

5. Conclusion

In this paper we have analyzed the relationship between credit constraints and firms' R&D investment behavior over the business cycle using a French firm-level panel data set over the period 1994–2004. We showed that: (i) R&D investment is countercyclical without credit constraints, but it becomes procyclical as firms face tighter credit constraints; (ii) this result is magnified for firms in sectors that depend more heavily upon external finance, or that are characterized by a low degree of asset tangibility; (iii) in more-credit-constrained firms, R&D investment falls during recessions but does not increase proportionally during upturns.

An important next step in this research program will be to study the effect of macroeconomic policy—both monetary and budgetary policies—on firms' R&D behavior over the business cycle. In particular, more countercyclical macroeconomic policies (e.g. with higher fiscal deficits or lower interest rates in downturns) should enhance R&D investments and productivity growth in firms that are more credit constrained and more dependent upon external finance. However, a systematic investigation of the effects of macroeconomic policies on firms' investment behavior is left for future research.

Appendix A

TABLE A.1. Variables description.

Variable	Description	Source
New bank loans	Total value of new bank loans	Centrale des Bilans, Banque de France
Payment incident	1 when the firm experienced at least one payment incident, 0 otherwise	Observatoire des entreprises, BdF
Δ Sales	$\text{Log}(\text{sales}) - \text{Log}(\text{sales}_{t-1})$	Fiben, BdF
Size	Number of employees	Fiben, BdF
Collateral	Sum of fixed and tangible assets	Fiben, BdF
Banking debt	Banking debt/ (Own financing + Market financing + Financial debt)	Fiben, BdF
Exchange rate	Real bilateral exchange rate	Penn World Tables
GDP	Yearly real GDP	Penn World Tables
Exports	Destination-specific export values	Balance of payments, BdF
R&D investment	Capitalized R&D expenditures	Fiben, BdF
Productivity	Total factor productivity (TFP)	Fiben, BdF

TABLE A.2. First-stage estimations.

Dep. var.	(a) $\Delta \text{Sales}(t)$	(b) $PI(t-1)$
$RER(t)$	-0.018*** (0.003)	0.003 (0.007)
$RER(t-1)$	-0.006* (0.003)	0.004 (0.007)
$RER(t-2)$	0.011** (0.003)	0.001 (0.007)
$GDP(t)$	0.004 (0.003)	0.005 (0.006)
$GDP(t-1)$	0.006** (0.003)	0.006 (0.006)
$GDP(t-2)$	-0.003 (0.003)	0.010 (0.006)
No. obs.	33763	33763
R^2	0.88	0.62

Notes: First-stage estimations of Table 4, column (b). Robust standard errors in parentheses. Estimations include sector $\times$ year dummies, firm fixed-effects, and the other instrumental variables.

* Significant at 10%; ** significant at 5%; *** significant at 1%.

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